Aula 4

Wednesday, March 11, 2026 8:02 AM

Minimização Cross-Entropy-Loss

$$L(w) = \frac{-1}{N} \cdot \sum_{n=1}^{N} y^{(n)} \ln \hat{p}_x(n) + (1 - y^{(n)}) \ln (1 - \hat{p}_x(n))$$

$$\hat{p}_x = h(x) = \sigma(w^t \tilde{x}) = \frac{1}{1 + e^{-w^t \tilde{x}}}$$

Derivada parcial

$$\frac{\partial}{\partial w_j} L(w) = \sum_{n=1}^{N} (\hat{p}_x(n) - y^{(n)}) x_j^{(n)}$$

Comentário

↳ Porque não usar MSE? Porque Cross-Entropy

- Explicação James D. McCaffrey

Classifier performance evaluation

Matriz de confusão

P → positivo

N → negativo

- TP → true positive $\hat{y} = 1$ $y = 1$
- FP → false positive $\hat{y} = 1$ $y = 0$
- FN → false negative $\hat{y} = 0$ $y = 1$
- TN → true positive $\hat{y} = 0$ $y = 0$

	Positivo	Negativo
Pos	TP	FP
Neg	FN	TN

$$\text{Precisão} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

1 → Valor ideal

$$y^{(i)} = \begin{cases} 1 & , h(x^i) \geq T \\ 0 & , \text{caso contrario} \end{cases}$$

Valor T influencia TP, FP, FN, TN